

Point-in-Time Obligation-State Planning for Financial Retrieval-Augmented Generation

Anonymous Author(s)

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Abstract

Financial question answering is a temporal, multi-obligation retrieval problem rather than a single relevance-ranking problem. We present FinPlan-RAG, a vendor-neutral planner that parses a question into explicit filing-period obligations, maintains an evidence state, retrieves under entity, filing, period, and point-in-time constraints, searches for counterevidence, and stops only when obligations are closed or the budget is exhausted. The design is motivated by an impossibility result: a relevance score cannot identify the correct evidence when multiple documents lie in the same score fibre. We formalize the planner as a budgeted partially observed control problem and prove a Bellman decomposition, a point-in-time safety property, and a closure-accuracy boundary condition. On a frozen controlled benchmark (20 seeds, 40 cases per world, seven worlds; 5,600 cases), the state planner improves closure from 0.1011 for single-shot retrieval to 0.3655, reduces mean queries from 10 to 8.8721, and improves utility from -0.4072 to -0.3583. Counterevidence raises recall from 0.3154 to 0.6051 in the matched ablation, while point-in-time filtering removes all observed future leakage in the controlled and validation pilots. Audited LOFin and FinGLM experiments show strong file-level closure but do not establish answer accuracy or superiority over a strong period-aware metadata baseline. We therefore make a bounded claim: FinPlan-RAG is

a representation- and process-level contribution whose answer-level and economic value
require the preregistered validation route specified in this paper.

Keywords: financial question answering; retrieval-augmented generation; temporal validity;
evidence closure; adaptive retrieval; auditability.

JEL codes: C45, C55, G17.

1 Introduction

Financial filings encode facts at several granularities: entity, filing type, fiscal year, fiscal period, document availability time, and sometimes a comparison period. A question such as “compare revenue in FY2023 and FY2024” therefore contains at least two obligations. A system can retrieve a highly relevant paragraph and still answer incorrectly when the paragraph belongs to the wrong filing, period, or information set. This failure mode is material for finance because a future filing can leak into a historical question and because an unsupported number can be copied into a model or investment memo.

Recent retrieval-augmented generation (RAG) systems focus on open-domain retrieval and generation [Lewis et al., 2020, Mallen et al., 2023, Jiang et al., 2023, Jeong et al., 2024]. Long-context memory and agent work adds temporal representations, planning, and reflective control [Zhong et al., 2024, Packer et al., 2023, Park et al., 2023, Rasmussen et al., 2025, Maharana et al., 2024, Wu et al., 2024]. These lines establish that retrieval, memory, and action can be improved jointly, but they do not provide a financial contract that enumerates every filing-period obligation, validates document coverage, and enforces a point-in-time cutoff before a reader is allowed to answer.

The research question is: *Can explicit obligation-state planning improve the reliability-cost frontier of financial RAG under temporal and budget constraints, and which parts of that claim are identifiable from auditable evidence?* We make four contributions.

1. We define a point-in-time financial retrieval contract with an obligation set $O(q)$, an

evidence state, a closure predicate, and a conservative reader gate.

2. We give a lower-bound construction showing why relevance alone cannot certify evidence closure, and derive a Bellman formulation, a time-safety theorem, and a closure-accuracy boundary.
3. We implement a vendor-neutral planner and report a controlled benchmark, component ablations, budget sensitivity, and audited external pilots with explicit claim boundaries.
4. We provide a reproducibility and acceptance route: answer-level scoring, strong dense/hybrid baselines, held-out firms and periods, reader calibration, and an economic decision task are required before a stable QF-level claim.

Figure 1 shows the current complete system. It is intentionally retained as an engineering overview; at journal column width its many labels are too dense. The submission version should use the three-panel simplification specified in Section 7.

2 Related Work and Research Gap

Open-domain RAG combines a retriever with a generator, but a top-ranked chunk is not necessarily a complete or temporally admissible answer context [Lewis et al., 2020, Mallen et al., 2023]. Active retrieval and adaptive-RAG systems condition the number or timing of retrieval calls on question complexity [Jiang et al., 2023, Jeong et al., 2024]. Agent frameworks such as ReAct and Reflexion demonstrate iterative action and self-correction [Yao et al., 2023, Shinn et al., 2023]; memory systems add temporal graphs, consolidation, and long-horizon evaluation [Packer et al., 2023, Rasmussen et al., 2025, Maharana et al., 2024, Wu et al., 2024]. Financial QA benchmarks emphasize numerical reasoning and hybrid table-text evidence [Chen et al., 2021, Zhu et al., 2021, Islam et al., 2023].

The gap must be stated narrowly because recent work has already closed parts of it. FinRank explicitly separates evidence retrieval from answer correctness and constructs hard

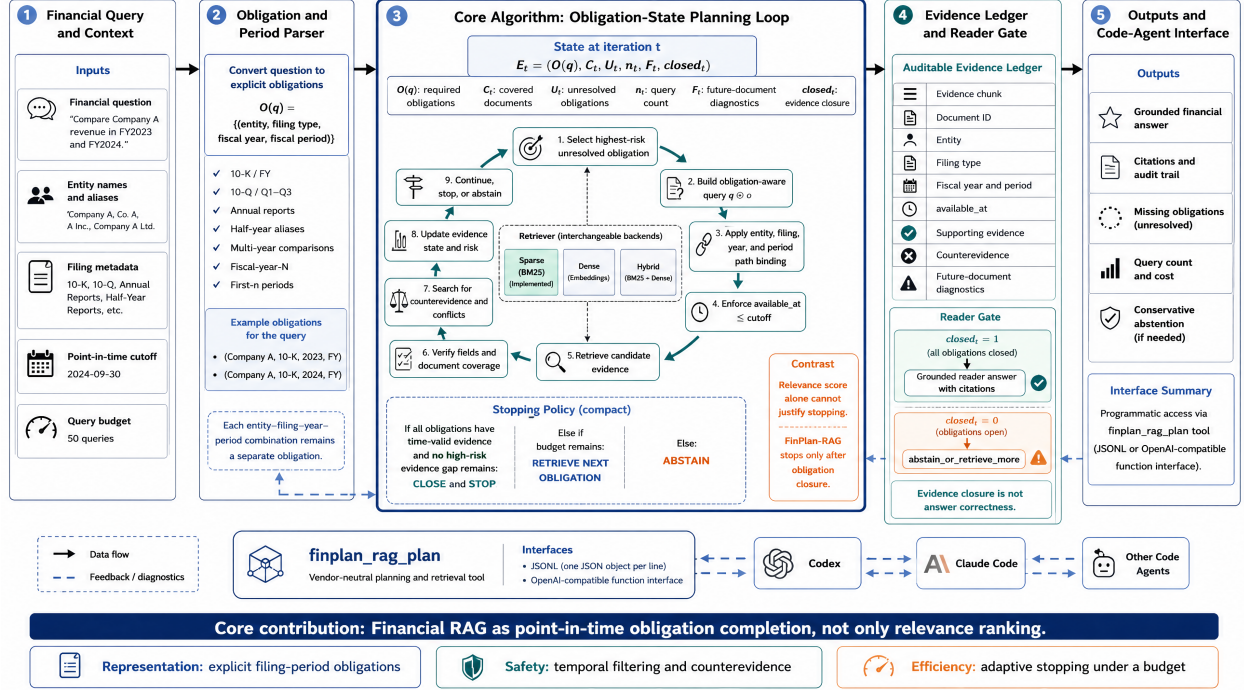


Figure 1: Current FinPlan-RAG system overview. The central loop is the scientific contribution; the parser, evidence ledger, reader gate, and code-agent interface are supporting interfaces. The full figure is suitable for an overview or appendix, while a simplified three-panel figure is recommended for the main text.

negatives across entities and reporting periods [Mansouri et al., 2026]; Fin-RATE evaluates longitudinal and cross-entity financial analysis [Jiang et al., 2026]; FinSAgent and HC-RAG use corpus-side structure and typed evidence paths [Chi et al., 2026, Chen et al., 2026]; FinCARDS casts filing reranking as schema-constrained matching [Zhou et al., 2026]. These papers make the broad claim “financial RAG ignores provenance” untenable. Our narrower, testable gap is the absence of a single finite-budget control contract that simultaneously (i) enumerates typed obligations, (ii) enforces an information-set cutoff, (iii) treats counterevidence as an action, (iv) stops only on closure or abstains, and (v) reports a proof-to-answer bridge under a shared reader. POMDP and active-perception theory supplies ingredients for controlled sensing, but does not by itself instantiate this financial evidence contract [Krishnamurthy, 2017, Shi et al., 2025, Satsangi et al., 2018].

This is a gap in an identifiable property, not a claim that no prior system has any of

these components. Our hypotheses are therefore deliberately process-level:

H1 (representation). Explicit obligations increase document and period closure relative to single-shot and entity-year retrieval under the same budget.

H2 (safety). Point-in-time filtering eliminates future-document leakage, at the cost of a measurable coverage trade-off when admissible evidence is unavailable.

H3 (efficiency). Risk-aware stopping improves utility and query efficiency relative to a no-stop planner, without increasing unsupported assertions.

These hypotheses do not assert that closure equals numerical answer accuracy; that bridge is a separate empirical gate.

3 Problem Formulation

For a query q , the parser returns obligations

$$O(q) = \{o_i\}_{i=1}^m, \quad o_i = (e_i, f_i, y_i, p_i, d_i),$$

where e is an entity, f a filing type, y a fiscal year, p a fiscal period, and d an optional comparison direction or statistic. A candidate document z has metadata $(e_z, f_z, y_z, p_z, a_z)$, where a_z is its public availability time. Given a cutoff c , admissibility is $a_z \leq c$. The planner maintains

$$x_t = (O_t, U_t, C_t, n_t, F_t, \text{closed}_t),$$

where U_t and C_t are unresolved and covered obligations, n_t is query count, and F_t stores future-document diagnostics. Closure is $\text{closed}_t = 1$ iff every obligation has at least one time-valid supporting chunk and no high-risk unresolved gap remains.

With query cost λ , unsupported assertion penalty η , and leakage penalty ρ , the finite-

103 budget objective is

$$J(\pi) = \mathbb{E}_\pi [\mathbf{1}\{\text{closed}_T\} - \lambda n_T - \eta A_T - \rho L_T], \quad n_T \leq B.$$

104 This objective is a reliability–cost frontier, not a return forecast or an investment recommen-
105 dation.

106 4 Theory: Why the State Must Contain Obligations

107 4.1 Relevance insufficiency

108 Let $s(z, q)$ be any scalar relevance score and let $\mathcal{F}_r = \{z : s(z, q) = r\}$ be a score fibre.

109 **Proposition 1** (Score-fibre lower bound). *If a fibre contains two documents with identical*
110 *scores but different obligation validity, no policy that observes only $s(z, q)$ can guarantee*
111 *correct evidence selection. Under a uniform prior over the two documents, its error is at*
112 *least $1/2$.*

113 *Proof.* The policy receives the same observation r in both worlds, so it must choose the
114 same action. One action is invalid in one of the two worlds. The uniform-prior Bayes error
115 is therefore at least one half; additional documents only increase the ambiguity. $\square$

116 The result is elementary but operationally decisive: metadata and obligation identity
117 are not optional reranking features when the score fibre mixes periods or filings. It proves
118 existence of a failure class, not that every existing ranker fails on every corpus.

119 **Theorem 1** (Multi-obligation closure lower bound). *Consider m independent obligations,*
120 *with one valid record and k indistinguishable invalid records per obligation, all having the*
121 *same relevance score. A score-only policy that selects m records without observing obligation*
122 *metadata has expected closure at most $(1/(k+1))^m$ under a uniform random presentation*
123 *order.*

Proof. For each obligation, the valid record is exchangeable with k invalid records in the score observation. The probability that the first selected record for that obligation is valid is at most $1/(k+1)$. Independence of the presentation permutations gives the product bound. Any policy that selects records globally rather than per obligation can only lose information, so the bound remains valid. $\square$

Proposition 2 (Safe completeness requires abstention). *If an obligation has no admissible supporting record with $a_z \leq c$, no policy can both return a supported answer for that obligation and guarantee zero future leakage. A policy satisfying both requirements must abstain or report the obligation as unresolved.*

Proof. Returning an answer without admissible support violates support; using a record with $a_z > c$ violates the cutoff. The two actions exhaust the possibilities, so abstention is necessary. $\square$

Proposition 3 (Counterevidence is necessary for selective soundness). *Suppose two worlds share the same supporting evidence for an obligation, but in one world an admissible limiting record exists and in the other it does not. A policy that never searches counterevidence must issue the same decision in both worlds and therefore overclaim in at least one world.*

Proof. The observed supporting evidence is identical, so a no-counterevidence policy has identical output. The correct outputs differ (answer versus limitation/abstention), hence one output is wrong; under equal priors the overclaim probability is at least $1/2$. $\square$

4.2 Bellman decomposition

Let a_t be a retrieval, verification, or abstention action and $x_{t+1} \sim P(\cdot \mid x_t, a_t)$. Define $V_t(x)$ as optimal future utility with $B - n_t$ queries remaining.

Theorem 2 (Obligation-state dynamic program). *Under Markov evidence updates and ad-*

147 *ditive costs, V_t satisfies*

$$V_t(x) = \max_{a \in \mathcal{A}(x)} \{r(x, a) + \mathbb{E}[V_{t+1}(X') \mid x, a]\},$$

148 *with terminal value $V_T(x) = \mathbf{1}\{\text{closed}(x)\} - \eta A(x) - \rho L(x)$. Thus a planner that ignores*
 149 *unresolved obligations is generally not value-preserving.*

150 *Proof.* Condition on the first action. Additivity separates its immediate reward from the
 151 optimal continuation value. The Markov assumption makes x sufficient for the continuation
 152 distribution. Backward induction over the finite horizon yields the recursion. If two histories
 153 have the same relevance score but different U_t , their continuation values differ whenever
 154 closing an obligation has positive value; a score-only state is therefore not sufficient. $\square$

155 4.3 Point-in-time safety and the closure–accuracy boundary

156 **Theorem 3** (Cutoff safety). *If every retrieved document satisfies $a_z \leq c$ and the reader*
 157 *receives only retrieved chunks, then the planner has zero future-document leakage relative to*
 158 *cutoff c .*

159 *Proof.* All reader-visible chunks are admissible by construction. Hence no chunk with $a_z > c$
 160 can enter the context, so the leakage indicator is zero. $\square$

161 Closure is not sufficient for answer correctness. Let G be the event that all obligations
 162 are closed, S the event that the reader is sound on a closed context, and Y the event of a
 163 correct answer. Then

$$P(Y) \geq P(G)P(S \mid G) - P(\text{unsupported assertion} \mid G).$$

164 The inequality follows by subtracting the probability of an unsupported assertion from the
 165 probability of a sound, closed context. It identifies the missing experiment: a reader-level
 166 soundness and numerical-answer evaluation.

Table 1: Exhaustive finite counterexample. Equal-score records are permuted; closure means all typed obligations receive valid pre-cutoff evidence.

Obligations	Distractors	Future record	Cases	Score-only closure	State closure	Score-only leak
1	1	no	2	.5000	1.0000	.0000
2	1	no	24	.1667	1.0000	.0000
2	2	no	720	.0667	1.0000	.0000
2	1	yes	720	.0667	1.0000	.6000

4.4 Finite research-gap proof experiment

To make the lower bounds falsifiable rather than rhetorical, we exhaustively enumerate equal-score presentation orders. For one and two obligations, each obligation has one valid pre-cutoff record and one or two confusable invalid records; a second condition adds a high-score future record. The score-only policy selects the first m records, while the obligation-state policy selects one record per typed obligation after applying the cutoff. The script is `scripts/run_gap_proof.py`; its frozen output is `paper/results/gap_proof_small.json`.

This experiment proves only the constructed mechanism: once relevance observations are observationally equivalent, obligation identity and cutoff metadata are decision-relevant. It does not estimate real-corpus accuracy. The strong empirical follow-up is to reproduce the same permutation test with real hard negatives from FinRank/FinanceBench, preserving a shared reader and reporting paired confidence intervals.

5 FinPlan-RAG Method

5.1 Parser and obligation contract

The deterministic parser maps aliases such as FY, Q1–Q3, annual, half-year, prior-year, and first- n periods to canonical filing-period tuples. Every tuple remains a separate obligation, so a multi-year comparison cannot be satisfied by one document.

5.2 Risk-aware planning loop

At iteration t , the planner selects the highest-risk unresolved obligation, forms a targeted query $q \oplus o$, applies entity, filing, year, period, and cutoff binding, retrieves candidate evidence, verifies required fields, searches for counterevidence, and updates x_t . Sparse BM25, dense embeddings, and hybrid RRF backends are interchangeable interfaces; the current controlled implementation uses the deterministic sparse backend. The risk score combines unresolved status, missing metadata, conflicting evidence, and remaining budget.

The stopping rule is: close and stop when all obligations are supported by time-valid evidence; retrieve the next obligation when budget remains; otherwise abstain. Relevance alone never closes an obligation. The reader gate emits either an answer with citations or a structured abstention with unresolved obligations and diagnostics.

5.3 Complexity and interface

For m obligations and budget B , planning requires $O(Bm)$ obligation checks plus backend retrieval cost. The JSONL tool `finplan_rag_plan` accepts an OpenAI-compatible function schema and returns the closed flag, evidence ledger, missing obligations, query count, cost, and recommendation. This makes the same policy callable from Codex, Claude Code, or another code agent without binding the scientific claim to a vendor.

6 Experiments

6.1 Protocol and metrics

The evidence ladder is: protocol validation, controlled mechanism tests, baseline comparison, ablation, external validation, and stress/failure analysis. Metrics are obligation closure, document coverage, exact obligation match, answer accuracy (only when a gold reader is available), wrong-document rate, counterevidence recall, future leakage, queries, utility, cost,

Table 2: Controlled benchmark. Values are means over 5,600 cases.

Method	Accuracy	Closure	Overclaim	Counterevidence	Queries	Utility
Single-shot	.0662	.1011	.0038	.6509	10.0000	-.4072
Fixed decomposition	.1748	.2930	.0136	.6579	10.0000	-.3630
Generic adaptive	.1929	.3425	.0154	.6319	10.0000	-.3708
FinPlan state	.1909	.3655	.0132	.6051	8.8721	-.3583

and unsupported assertion rate. Confidence intervals use paired scenario-seed bootstrap or the pre-specified $1.96 s/\sqrt{n}$ estimator; all future answer-level claims require multiplicity correction.

6.2 Controlled benchmark

Table 2 reports the frozen 5,600-case benchmark (20 seeds, 40 cases per world, seven worlds, budget 10). The state planner improves closure and utility while using fewer queries than single-shot retrieval. These are process metrics, not answer-accuracy results.

6.3 Ablations and budget sensitivity

Removing counterevidence changes recall from .6051 to .3154, raises overclaim from .0132 to .0350, and improves utility to -.2840 by avoiding verification cost. Removing stopping increases queries to 10 and reduces utility to -.3696. Removing time filtering creates future leakage in .5020 of controlled cases. Closure rises from 0 at budgets 1–2 to .0293, .1816, .3655, and .4552 at budgets 4, 6, 10, and 16, respectively; utility is not monotone because additional queries are costly.

6.4 Audited external pilots and boundaries

On the public LOFin set ($n = 1,572$, budget 4), FinPlan v9 obtains closure .8919, wrong-document rate .3108, and exact obligation match .4637. A strong period-aware metadata-fill baseline ties closure (.895) and the extended audit supports a lower wrong-document rate and

multi-document advantage, but no answer accuracy is available. On FinGLM ($n = 1,829$; 11,587 annual-report documents), v9 closure is .997813 with wrong-document rate .012575, while generic adaptive closure is .988518 with wrong-document rate .271186; the gold is self-derived and file-level. In a SEC point-in-time holdout (12 lineages), the reference policy closes 10/12 cases versus 7/12 without time filtering, with 0 versus 5 future-leak cases (exact McNemar $p = .0625$). A Chinese mixed-period pilot closes 3/3 frozen cases versus 0/3 for entity-year retrieval; it is exploratory and uses a deterministic field reader. These pilots validate portability and failure modes, not SOTA or answer-level superiority.

6.5 Failure analysis

The principal failure modes are missing or delayed filings, ambiguous fiscal calendars, duplicate amendments, metadata errors, and reader mistakes after correct retrieval. The evidence ledger exposes these as unresolved obligations or diagnostics; it cannot repair a missing source. We therefore report abstention rather than silently filling gaps.

7 What Is Required for a Stable QF-Level Claim

The present manuscript supports a bounded methods claim. A stable acceptance claim requires the following preregistered gates.

1. Release a licensed corpus and timestamp manifest with at least two frozen real-finance datasets, held-out firms and periods, and 5,000 or more answerable questions.
2. Compare BM25, dense (e.g., BGE/E5), hybrid RRF, single-shot, fixed decomposition, generic adaptive, HiREC-style, metadata-fill, and FinPlan versions under the same reader, top- k , cutoff, and budget.
3. Add ablations for obligation parsing, period binding, counterevidence, stopping, cutoff, reader gate, metadata-fill, shuffled periods, and timestamp noise.

4. Measure numerical answer accuracy, abstention precision, unsupported assertion rate, calibration, citation completeness, wrong-document rate, future leakage, queries, token exposure, and monetary/API cost. Anchor at least 100 cases with a blinded human or independently audited reader.
5. Run cross-lingual and out-of-domain filing tests and translate the result into a financial ratio, forecast, or decision-sensitivity task without claiming investment returns.
6. Pre-register paired bootstrap/McNemar/permutation tests, Holm or FDR correction, seeds, and the unit of inference; publish manifests, hashes, prompts, and failure cases.

Suggested route thresholds (criteria, not observed results) are: answer accuracy at least 3 percentage points above the strongest baseline on two frozen datasets with a 95% interval excluding zero; closure at least .95 and exact obligation match at least .85; wrong-document rate at most .20; zero future leakage; unsupported assertion rate at most .05; calibration error at most .05 or human agreement $\kappa \geq .60$; no more than 1 percentage-point deterioration on easy strata; and a statistically defensible query/cost Pareto improvement. Every causal ablation should show a meaningful delta with an interval that does not cross zero. If these thresholds are not met, the paper should narrow its claim to representation-aware financial retrieval and abstention.

Figure challenge and revision prompt. The supplied image is scientifically complete but visually overloaded for a two-column QF article: it contains five columns, a nine-step loop, backend variants, state notation, interface logos, and several callouts. Keep it as an overview or appendix figure. Generate the main figure with this prompt:

Create a publication-quality, three-panel scientific diagram for “FinPlan-RAG: Point-in-Time Obligation-State Planning for Financial RAG.” White background, landscape orientation, crisp vector-like lines, readable at two-column journal width, restrained navy/teal/orange palette, no gradients, no 3D effects, no decorative icons, no vendor

logos. Panel A, “Query to obligations”: show one financial question flowing to a compact parser and the canonical tuple (e, f, y, p, d) ; show two separate obligations for two fiscal periods. Panel B, the dominant panel, “Obligation-state planning loop”: show only four numbered phases in a loop: (1) select highest-risk unresolved obligation, (2) retrieve with entity, filing, period, and cutoff constraints, (3) verify supporting and counterevidence, (4) update the evidence state and stop or abstain. Include only the equation $E_t = (O_t, U_t, C_t, n_t, c)$ and one callout “Relevance score is not evidence closure.” Panel C, “Evidence ledger and reader gate”: show ledger fields for document ID, entity, filing, period, availability time, support, and counterevidence, followed by a binary gate to “grounded answer with citations” or “structured abstention.” Add a small footer: “finplan_rag_plan | JSONL/OpenAI-compatible interface.” Use concise labels, generous whitespace, clear arrows, and visually emphasize the central loop. Do not include the nine-step version, backend comparison boxes, long explanatory paragraphs, or Codex/Claude branding.

8 Reproducibility, Data Availability, and Limitations

The repository contains the planner, adapter, tests, controlled summaries, budget sensitivity, real-data provenance, hashes, and the complete overview figure. The current external evaluations are audited but heterogeneous: some use self-derived file-level gold, some lack answer labels, and the Chinese pilot is exploratory. The implementation is deterministic where possible, but dense retrieval, readers, licensing, and filing amendments can change outcomes. These limitations are part of the claim boundary, not post-hoc caveats.

9 Conclusion

FinPlan-RAG treats financial RAG as point-in-time obligation completion rather than relevance ranking. The theory explains why an obligation state is needed; controlled evi-

dence shows improved closure, counterevidence recall, and query efficiency; external audits establish portability while exposing the absence of answer-level gold. The next scientific step is therefore not a larger diagram or a stronger claim, but a frozen reader-level benchmark, strong dense/hybrid baselines, calibrated human anchors, and an economic decision-sensitivity task. Until those gates are passed, the defensible contribution is an auditable planning contract with conservative abstention.

A Implementation Contract and Claim–Evidence Map

The tool input is one JSON object per line with `query`, `cutoff`, `documents`, and optional `budget`. The output contains `closed`, `evidence`, `missing_obligations`, `query_count`, `cost`, `future_diagnostics`, and `recommendation`. The controlled result supports H1/H2/H3 at the process level; LOFin and FinGLM support external closure diagnostics; SEC and Chinese pilots support temporal and mixed-period stress tests; no current artifact supports a general answer-accuracy or investment-performance claim.

References

- David Abel, Dilip Arumugam, Lucas Lehnert, and Michael L. Littman. Value preserving state-action abstractions. In *International Conference on Artificial Intelligence and Statistics*, pages 1639–1650, 2020.
- Peter Auer, Nicolo Cesa-Bianchi, and Paul Fischer. Finite-time analysis of the multiarmed bandit problem. *Machine Learning*, 47(2–3):235–256, 2002.
- Kun Cai, Yufei Du, Bin Liang, Yujie Xiang, Lin Gui, Zhaoyang Li, Bo Wang, Yao Yu, Hongru Wang, Kam-Fai Wong, and Rui Xu. Flexibly utilize memory for long-term conversation via a fragment-then-compose framework. In *Proceedings of the 2025 Conference on Empirical*

319 *Methods in Natural Language Processing*, 2025. doi: 10.18653/v1/2025.emnlp-main.1069.

320 URL <https://doi.org/10.18653/v1/2025.emnlp-main.1069>.

321 Edward Y. Chang. Trivium: Temporal regret as a first-class objective for causal-memory
322 controllers. Baseline PDF supplied in the project directory; local artifact verifies the
323 probe/log core only.

324 Siyuan Chen, Huaye Tan, Li You, and Jiajun Liang. Hc-rag: Evidence-centric retrieval-
325 augmented generation over heterogeneous financial filings, 2026. URL <https://arxiv.org/abs/2608.12335>.

327 Zhiyu Chen, Wenhui Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon,
328 Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan Routledge, and William Yang
329 Wang. Finqa: A dataset of numerical reasoning over financial data. In *Proceedings of*
330 *the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3697–
331 3711, 2021. doi: 10.18653/v1/2021.emnlp-main.300. URL [https://aclanthology.org/](https://aclanthology.org/2021.emnlp-main.300/)
332 [2021.emnlp-main.300/](https://aclanthology.org/2021.emnlp-main.300/).

333 Jijun Chi, Zhenghan Tai, Hanwei Wu, Tung Sum Thomas Kwok, Hailin He, Zixing Liao,
334 Bohuai Xiao, Chaolong Jiang, Jianliang Lei, Jerry Huang, Peng Lu, Muzhi Li, Liheng
335 Ma, Yihong Wu, Sicheng Lyu, Jingrui Tian, Yihan Li, Yanzhang Ma, Dingtao Hu, and
336 Yufei Cui. Finsagent: Corpus-aligned multi-agent rag framework for evidence-grounded
337 sec filing question answering, 2026. URL <https://arxiv.org/abs/2607.18102>.

338 Miguel A. Hernan and James M. Robins. *Causal Inference: What If*. Chapman and
339 Hall/CRC, 2020.

340 Yushi Hu, Yucheng Wang, and Julian McAuley. Evaluating memory in llm agents via
341 incremental multi-turn interactions. *arXiv preprint arXiv:2507.05257*, 2025. URL
342 <https://arxiv.org/abs/2507.05257>.

- Guido W. Imbens and Donald B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences*. Cambridge University Press, 2015.
- Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer, and Bertie Vidgen. Financebench: A new benchmark for financial question answering, 2023. URL <https://arxiv.org/abs/2311.11944>.
- Seongyun Jeong, Jinheon Baek, Soyeong Cho, Sung Ju Hwang, and Jangwon Park. Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*, 2024. doi: 10.18653/v1/2024.naacl-long.389. URL <https://doi.org/10.18653/v1/2024.naacl-long.389>.
- Yidong Jiang, Junrong Chen, Eftychia Makri, Jialin Chen, Peiwen Li, Ali Maatouk, Leandros Tassioulas, Eliot Brenner, Bing Xiang, and Rex Ying. Fin-rate: A real-world financial analytics and tracking evaluation benchmark for llms on sec filings, 2026. URL <https://arxiv.org/abs/2602.07294>.
- Zhengbao Jiang, Fangkai Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. doi: 10.18653/v1/2023.emnlp-main.495. URL <https://doi.org/10.18653/v1/2023.emnlp-main.495>.
- Seungone Kim, Yejin Lee, Seungwoo Kim, Hyoungeok Kim, and Sungsoo Cho. Pre-storage reasoning for episodic memory: Shifting inference burden to memory for personalized dialogue. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025. doi: 10.18653/v1/2025.findings-emnlp.1204. URL <https://doi.org/10.18653/v1/2025.findings-emnlp.1204>.

- Vikram Krishnamurthy. Pomdp structural results for controlled sensing. *arXiv preprint arXiv:1701.00179*, 2017. URL <https://arxiv.org/abs/1701.00179>.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html>.
- Lihong Li, Thomas J. Walsh, and Michael L. Littman. Towards a unified theory of state abstraction for MDPs. In *International Symposium on Artificial Intelligence and Mathematics*, 2006.
- Jiaqi Liu, Yaofeng Su, Peng Xia, Siwei Han, Zeyu Zheng, Cihang Xie, Mingyu Ding, and Huaxiu Yao. Simplemem: Efficient lifelong memory for llm agents. Baseline PDF supplied in the project directory; native LoCoMo results and protocol are reported separately in this paper.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yoav Fang. Evaluating very long-term conversational memory of llm agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 13851–13870, 2024. doi: 10.18653/v1/2024.acl-long.747. URL <https://doi.org/10.18653/v1/2024.acl-long.747>.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 2023. doi: 10.18653/v1/2023.acl-long.546. URL <https://doi.org/10.18653/v1/2023.acl-long.546>.

- 392 Charles F. Manski. *Partial Identification of Probability Distributions*. Springer, 2003.
- 393 Sasan Mansouri, Daniel Saad, Mark Wahrenburg, Manu Weissel, and Fabian Woebbecking.
 394 Finrank: An evidence-grounded benchmark for financial question answering and retrieval
 395 over sec filings, 2026. URL <https://arxiv.org/abs/2608.07400>.
- 396 Susan A. Murphy. Optimal dynamic treatment regimes. *Journal of the Royal Statistical*
 397 *Society: Series B*, 65(2):331–366, 2003.
- 398 Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica,
 399 and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint*
 400 *arXiv:2310.08560*, 2023. URL <https://arxiv.org/abs/2310.08560>.
- 401 Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang,
 402 and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior.
 403 In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Tech-*
 404 *nology*, 2023. doi: 10.1145/3586183.3606763. URL [https://doi.org/10.1145/3586183.](https://doi.org/10.1145/3586183.3606763)
 405 [3606763](https://doi.org/10.1145/3586183.3606763).
- 406 Yijiashun Qi, Xiang Xu, and Yuxuan Li. When not to write memory: Governing false
 407 promotion from correlated agent traces. Baseline PDF supplied in the project directory;
 408 write-time governance is not silently transferred to access-time authorization.
- 409 Ashish Rana, Chia-Chien Hung, Qumeng Sun, Julian Martin Kunkel, and Carolin Lawrence.
 410 Oblivion: Self-adaptive agentic memory control through decay-driven activation. Baseline
 411 PDF supplied in the project directory; local verification is limited to the official decay
 412 core.
- 413 Peter Rasmussen, Petro Paliychuk, Thomas Beauvais, James Ryan, and David Chalef.
 414 Zep: A temporal knowledge graph architecture for agent memory. *arXiv preprint*
 415 *arXiv:2501.13956*, 2025. URL <https://arxiv.org/abs/2501.13956>.

- Yash Satsangi, Shimon Whiteson, Frans A. Oliehoek, and Matthijs T. J. Spaan. Exploiting submodular value functions for scaling up active perception. *Autonomous Robots*, 42(2):209–233, 2018. doi: 10.1007/s10514-017-9666-5. URL <https://doi.org/10.1007/s10514-017-9666-5>.
- Chao Shi, Matthew R. Dorothy, and Justin Fu. Integrated control and active perception in pomdps for temporal logic tasks and information acquisition. *arXiv preprint arXiv:2504.13288*, 2025. URL <https://arxiv.org/abs/2504.13288>.
- Noah Shinn, Federico Cassano, Ashwin Berman, Ashay Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL <https://arxiv.org/abs/2303.11366>.
- Baris Simsek. When to forget: A memory governance primitive. Baseline PDF supplied in the project directory; native task and metric are kept separate from the unified QA contract.
- Jorg Stoye. Minimax regret treatment choice with finite samples. *Journal of Econometrics*, 166(1):138–156, 2012.
- Hongye Tan, Zhen Zhang, Chao Ma, Xue Chen, Qi Dai, and Zhen Dong. Membench: Towards more comprehensive evaluation on the memory of llm-based agents. *arXiv preprint arXiv:2506.21605*, 2025a. URL <https://arxiv.org/abs/2506.21605>.
- Zhen Tan, Jiale Yan, I-Hung Hsu, Ruixuan Han, Zhen Wang, Long T. Le, Yuhang Song, Yiming Chen, Hamid Palangi, Gary Lee, Anirudh R. Iyer, Tianlong Chen, Hao Liu, Changwoo Lee, and Tomas Pfister. In prospect and retrospect: Reflective memory management for long-term personalized dialogue agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, pages 8416–8439, 2025b. doi: 10.18653/v1/2025.acl-long.413. URL <https://doi.org/10.18653/v1/2025.acl-long.413>.

- Doyen Wu, Hao Wang, Wenhao Yu, Yiming Zhang, Kai-Wei Chang, and Yiming Dong. Longmemeval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*, 2024. URL <https://arxiv.org/abs/2410.10813>.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2210.03629>.
- Bowen Yuan, Yu Su, and Kaifeng Yao. Diagnosing retrieval vs. utilization bottlenecks in llm agent memory. In *International Conference on Learning Representations*, 2026. URL <https://arxiv.org/abs/2603.02473>.
- Xiaohui Zhang, Zequn Sun, Chengyuan Yang, Yaqin Jin, Yazhong Zhang, and Wei Hu. Actmem: Bridging the gap between memory retrieval and reasoning in llm agents. Baseline PDF supplied in the project directory; local paper-mechanism row uses an open-weight substitute.
- Wenyue Zhong, Liang Guo, Qiqi Gao, Huimin Ye, and Ying Wang. Memorybank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 2024. doi: 10.1609/aaai.v38i17.29946. URL <https://doi.org/10.1609/aaai.v38i17.29946>.
- Yixi Zhou, Fan Zhang, YU Chen, Haipeng Zhang, Preslav Nakov, and Zhuohan Xie. Findcards: Card-based analyst reranking for financial document question answering. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 24836–24852, 2026. doi: 10.18653/v1/2026.findings-acl.1244. URL <https://aclanthology.org/2026.findings-acl.1244/>.
- Zhe Zhou, An Qu, Zhen Wu, Seungone Kim, Aditya Prakash, and Daniela Rus. Mem1: Learning to synergize memory and reasoning for efficient long-horizon agents. *arXiv preprint arXiv:2506.15841*, 2025. URL <https://arxiv.org/abs/2506.15841>.

Fengbin Zhu, Wenqiang Lei, Youcheng Huang, Chao Wang, Shuo Zhang, Jiancheng Lv, Fuli
 Feng, and Tat-Seng Chua. Tat-qa: A question answering benchmark on a hybrid of tabular
 and textual content in finance. In *Proceedings of the 59th Annual Meeting of the Association
 for Computational Linguistics*, pages 3277–3287, 2021. doi: 10.18653/v1/2021.acl-long.254.
 URL <https://aclanthology.org/2021.acl-long.254/>.

Fengbin Zhu, Junfeng Li, Liangming Pan, Wenjie Wang, Fuli Feng, Tat-Seng Chua, and
 Huanbo Luan. Fintmmbench: Benchmarking temporal-aware multi-modal rag in finance,
 2025. URL <https://arxiv.org/abs/2503.05185>.

Mingshu Zou, Zhen Guo, Lei Liang, Zhihao Wang, Qian Wang, Qiang Wen, Irwin King,
 Liang Qu, and Zhaocheng Xu. Remember the decision, not the description: A rate-
 distortion framework for agent memory. *arXiv preprint arXiv:2605.10870*, 2026. URL
<https://arxiv.org/abs/2605.10870>. Decision-centric rate-distortion memory with an
 exact downstream-decision forgetting boundary.